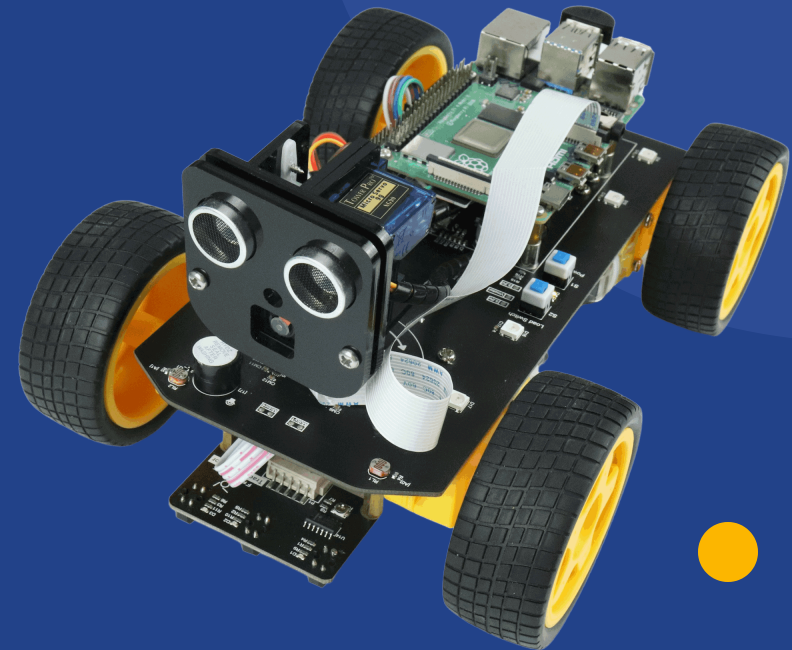




LESSON 0 – INITIAL SETUP

Machine Learning with Raspberry Pi & Smart Car

Before we write code, we prepare our robot — build the kit, flash the Pi, reach it remotely, and install the course code.



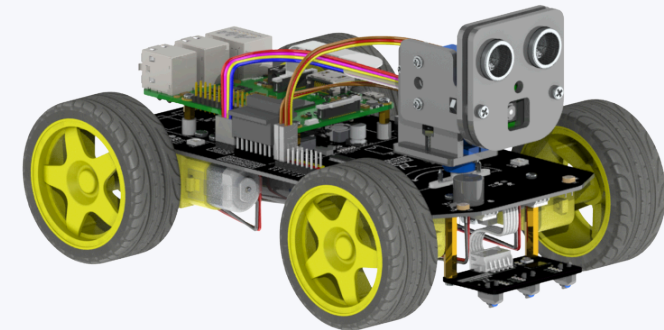


LESSON 0 – INITIAL SETUP

Welcome – today is all setup

Before we write code, we prepare the robot: the kit, the Raspberry Pi, and the course code.

- **Welcome to the course!** Before we write code, we prepare our robot.
- Today is **all setup** – the kit, the **Raspberry Pi** (the car's brain), and the course code.
- It's the longest, most technical session – but you only do it **once**.





LESSON 0 – INITIAL SETUP

By the end of today

A working, reachable Pi with the course code installed – and a ready-to-go computer.

- 1 Know the **parts** of your kit and how to build the car
- 2 Handle **batteries safely** and power the Pi correctly
- 3 **Flash** Raspberry Pi OS and boot the Pi
- 4 Reach the Pi from your computer with **SSH** and **VNC**
- 5 **Install the course code** and confirm it runs



The finish line: a Pi you can reach, with the code installed.



LESSON 0 – INITIAL SETUP

Two references you'll use all course

Keep these two PDFs handy – they live in the repo's `Resources/` folder.



[Tutorial.pdf](#)

Building the car and general kit guidance.



[About_Battery.pdf](#)

Choosing and handling batteries safely.

NOTE The code has been updated since the Tutorial was written – trust the **lesson decks** for code.

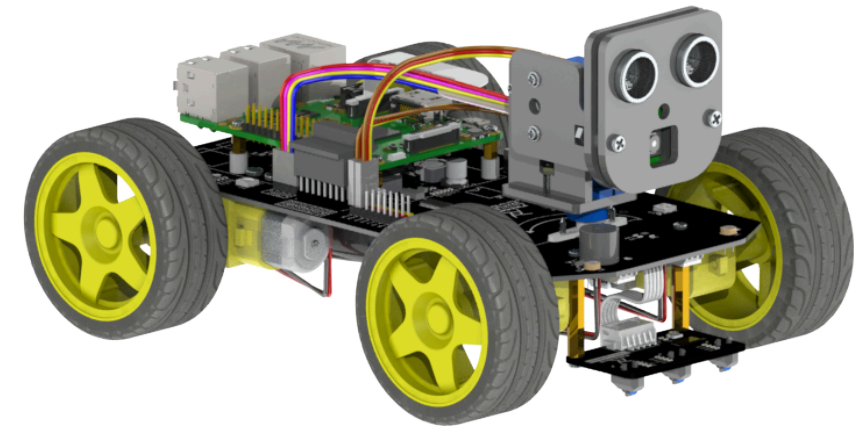


LESSON 0 – INITIAL SETUP

What's in the box

Your kit has five groups of parts – lay them out in these groups before we start.

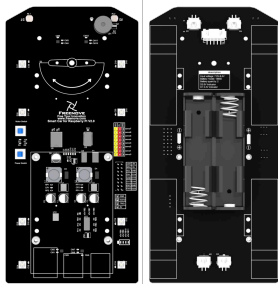
- **1 • Car board** + acrylic frame
- **2 • Transmission** parts (motors, wheels)
- **3 • Electronic** parts (sensors, camera, cables)
- **4 • Tools**
- **5 • Power** (batteries, charger, cables)



LESSON 0 – INITIAL SETUP • THE KIT

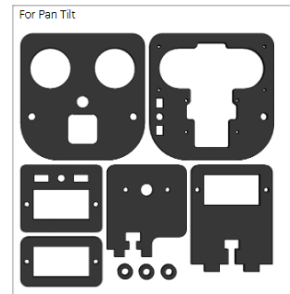
The car board & frame

The base the whole robot is built on.



Smart car board

The main circuit board the Pi sits on.



Acrylic parts

The layered frame and chassis.



Screw set

Holds it all together.



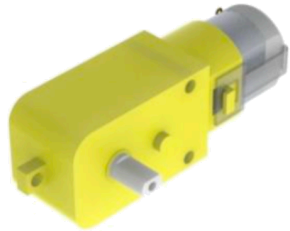
The acrylic may have a **protective paper film** to peel – it's not scratched.



LESSON 0 – INITIAL SETUP • THE KIT

Transmission: motors & wheels

These move the car – the parts you'll meet again in Lessons 1 and 2.



DC motors

Drive the four wheels.



Motor brackets

Hold motors to the frame.



Servo motor

Turns the camera head.



Wheels

Push onto the motor shafts.

LESSON 0 – INITIAL SETUP • THE KIT

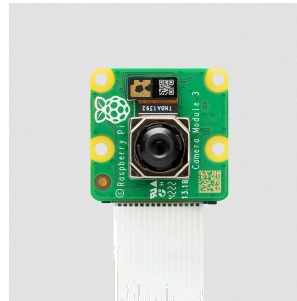
Electronic parts: sensors

Each sensor maps to a lesson that uses it.



Ultrasonic sensor

Distance – Lesson 4.



Camera module

Video – Lessons 6-8.



Line-tracker module

IR line sensors – Lesson 5.



LESSON 0 – INITIAL SETUP • THE KIT

Electronic parts: boards & cables

The pieces that wire the sensors to the Pi.



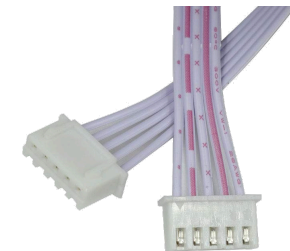
Connection board

Links sensors to the Pi.



Jumper wires

Wiring it together.



5-pin cable

Connects the modules.

LESSON 0 – INITIAL SETUP • THE KIT

Tools

A few small tools handle the whole build.



Screwdriver (3 mm)

For assembly.



Black tape

Keep it – it makes a line track in Lesson 5.



Wire tidy

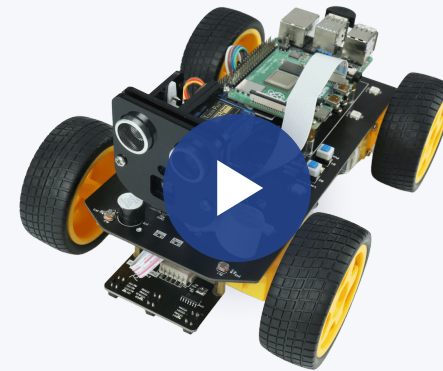
Keep cables neat.

LESSON 0 – INITIAL SETUP

Building the car

We recommend **watching the build video** and following [Resources/Tutorial.pdf](#) side by side.

- **Watch the build video** → and follow along – it walks the whole setup, start to finish.
- Open [Resources/Tutorial.pdf](#) for the step-by-step photos.
- Build the frame, mount **motors + wheels**, attach the **head/camera**, and seat the **Pi** on the board.
- Take your time and keep **screws organized**.



[Watch the smart car build video](#)



LESSON 0 – INITIAL SETUP • POWER

Power: what you'll use

During setup we power the Pi from the wall; the batteries are for driving later.



18650 batteries

Power the motors and the mobile Pi.



Battery charger

Recharge the 18650 cells.



AC power adapter

Powers the Pi directly during setup.



You'll also need the right **USB cable** for your Pi model – more on that next.



LESSON 0 – INITIAL SETUP • POWER

The 18650 batteries

The car uses 18650 rechargeable lithium cells – quality matters.

- The car uses **18650** rechargeable lithium cells.
- **Buy quality cells** – cheap or counterfeit ones can be unsafe and perform poorly.
- A reputable source: 18650battery.com
- Read [About_Battery.pdf](#) before charging.





LESSON 0 – INITIAL SETUP • POWER

Battery safety

Treat lithium batteries with respect – read this carefully.



Use the **correct type** for the car (see [About_Battery.pdf](#)).



Insert with the right **polarity** (+ / –), and **never short** the terminals – keep cells away from coins, keys, metal.



Don't use damaged cells – dented, leaking, or wrapper torn.



Charge with the **proper charger**; don't leave it charging unattended.



Keep away from **heat and water**; store in a cool, dry place.



When in doubt

Set a questionable cell aside and don't use it.



LESSON 0 – INITIAL SETUP • POWER

Charging the batteries

Charge early so the cells are ready when you want to drive.

- Use the included or recommended **charger**.
- Charge cells **fully** before the first drive.
- Watch **orientation** in the charger (+ / -).
- **Unplug when done** – don't overcharge or leave unattended.



TIP Most "car won't drive" problems are simply **weak batteries**.



LESSON 0 – INITIAL SETUP • POWER

Powering the Raspberry Pi

Use a proper power supply for your model – not a PC USB port.

Pi 5

5V / 5A

25W

Pi 4

5V / 3A

15W

Pi 3

5V / 2.5A

12W

- Use a **quality AC adapter and cable** – **not** a PC USB port.
- Underpowering causes a **lightning-bolt warning** and the Pi may not boot.



USB-C

Pi 4 & Pi 5



micro-USB

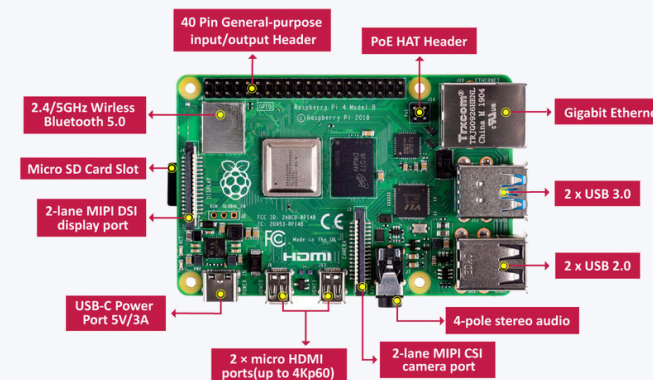
Pi 3

LESSON 0 – INITIAL SETUP • CHOOSING A PI

Which Raspberry Pi? – Pi 4 or 5

The course runs on a Pi 3B+, 4, or 5 – the Pi 4 is the one we recommend.

- ★ **Pi 4 recommended** – fast enough for the camera and vision lessons.
- Pi 4 and Pi 5 both use **USB-C** power.
- **Pi 5 works too** – but its camera port is different, so the kit's ribbon camera **won't connect** and the camera lessons (6–8) won't run on it.



Raspberry Pi 4 / 5

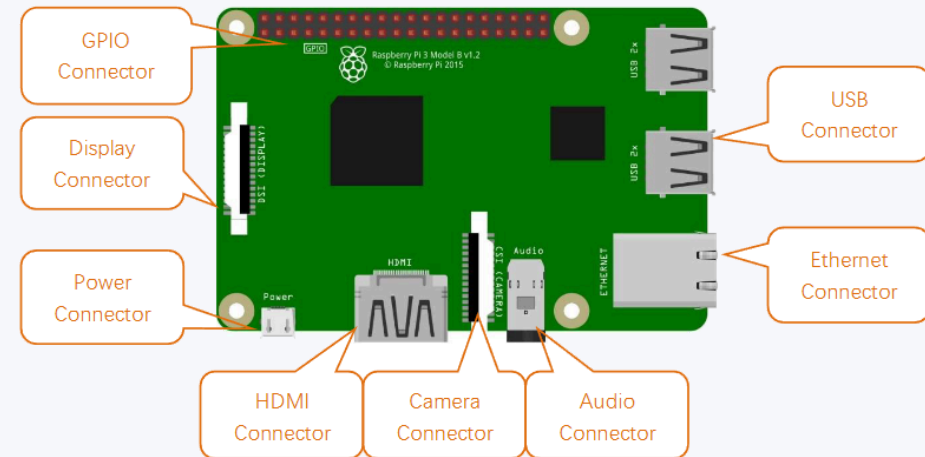
USB-C power

LESSON 0 – INITIAL SETUP • CHOOSING A PI

Which Raspberry Pi? – Pi 3B+

Also a great choice – it has the standard camera connector the kit needs.

- ★ **Standard camera connector** – the kit's ribbon camera plugs straight in.
- Uses **micro-USB** power – a different cable from the Pi 4 / 5.
- A little slower for the vision lessons (7–8), but everything still runs – just expect lower frame rates.



Raspberry Pi 3B+

micro-USB power



LESSON 0 – INITIAL SETUP • FLASHING THE PI

What you need to set up the Pi

Gather these before we start flashing.

- A **microSD card** + a card reader/adaptor.
- A **computer** (Windows / macOS / Linux) to flash the card.
- Your **Wi-Fi** name and password.
- The right **power adapter** for your Pi.



Have your Wi-Fi name & password ready

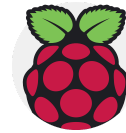


LESSON 0 – INITIAL SETUP • FLASHING THE PI

Step 1: install Raspberry Pi Imager

On **your computer**, download and install the free Imager tool.

- Download from raspberrypi.com/software.
- It's **free** and flashes Raspberry Pi OS onto the microSD card.
- Insert your **microSD card** (via the reader) when ready.



GRAB THE VERSION FOR YOUR OS





LESSON 0 – INITIAL SETUP • FLASHING THE PI

Step 2: choose device, OS & storage

Open Imager and pick three things.

1

Device

Your Pi model.

2

Operating system

Raspberry Pi OS (**32-bit**) – the Bookworm image, under "Raspberry Pi OS (other)". **Not** 64-bit.

3

Storage

Your microSD card – double-check you pick the SD card!



Flashing **erases** the chosen drive – confirm it's the SD card, not a USB drive.



64-bit *may* work if the course code has been updated since this video – feel free to test it. As of this video, **32-bit Bookworm is the stable, known-good version.**



LESSON 0 – INITIAL SETUP • FLASHING THE PI

Step 3: pre-configure (the important part!)

Before writing, open **OS customisation / Edit settings** and fill these in.

- **Hostname** – e.g. `mypi` → reachable as `mypi.local`.
- **Username & password** – pick something memorable.
- **Wi-Fi** name + password (+ your country).
- **Locale / time zone**.
- On the **Services** tab: **enable SSH** (password authentication).



Enable SSH + Wi-Fi here

This is what makes headless (no-monitor) setup possible –
and saves huge pain later.

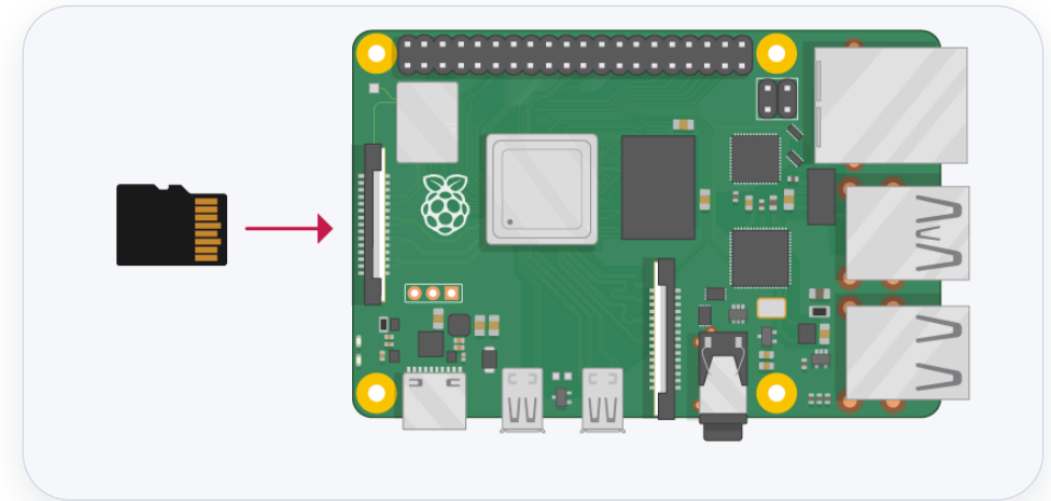


LESSON 0 – INITIAL SETUP • FLASHING THE PI

Step 4: write & first boot

Write the card, move it to the Pi, and give the first boot a moment.

- Click **Write** and wait for it to finish (and verify).
- **Eject** the card, put it into the Pi, connect power.
- The first boot takes a couple of minutes – **give it time**.



Don't pull power mid-boot – the Pi joins your Wi-Fi automatically using the Step-3 settings.



LESSON 0 – INITIAL SETUP • REACHING THE PI

Two ways to reach your Pi

Most setups are headless – no monitor on the Pi. We reach it from our computer two ways.



SSH

A text **terminal** on the Pi.



VNC

The Pi's full **desktop** in a window.

ORDER SSH first (already enabled from Step 3), then turn on VNC through SSH.



LESSON 0 – INITIAL SETUP • REACHING THE PI

Find your Pi

Reach the Pi by hostname or IP address.

- **Hostname:** `mypi.local` – the name you set in Step 3.
- **IP:** e.g. `192.168.1.50`.
- **Hostnames are handy** – if the IP changes, the name still works.

BY NAME

`mypi.local`

BY IP – RUN ON THE PI

`hostname -I`



LESSON 0 – INITIAL SETUP • REACHING THE PI

SSH into the Pi

From your computer's terminal, connect with the username + hostname you set.

- Say **yes** to the first-time fingerprint prompt, then enter your password.
- You're now typing commands **on the Pi**.
- On Windows, use **PowerShell / Terminal** – `ssh` is built in.

```
your computer . terminal

ssh <username>@<hostname>.local

# example:
ssh pi@mypi.local

# if .local fails, use the IP:
ssh pi@192.168.1.50
```



LESSON 0 – INITIAL SETUP • REACHING THE PI

Enable VNC on the Pi

Over your SSH connection, open the config tool and turn VNC on.

- Run `sudo raspi-config` over SSH.
- Go to **Interface Options** → **VNC** → **Enable**.
- The setup scripts handle **I2C** and the **camera** – VNC is the manual step.
- **Finish** and reboot if asked.

the Pi · over SSH

```
sudo raspi-config
```

```
# then choose:
```

```
#   Interface Options
```

```
#       → VNC
```

```
#           → Enable
```



LESSON 0 – INITIAL SETUP • REACHING THE PI

Install VNC Viewer on your computer

Download the free viewer for your operating system.

- Download **VNC Viewer** from realvnc.com.
- It's **free** – pick your OS.
- Newer versions may ask you to **create an account**.



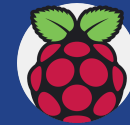


LESSON 0 – INITIAL SETUP • REACHING THE PI

Connect with VNC Viewer

Enter the Pi's address, log in, and the desktop appears.

- Open VNC Viewer and enter the Pi's **hostname or IP**: `mypi.local` (or `192.168.1.50`).
- Log in with your Pi **username & password**.
- The Pi's **desktop** appears in a window – you're in!



First connection?

It may warn about an identity / encryption check – accept to continue.

LESSON 0 – INITIAL SETUP • REACHING THE PI

Optional: Raspberry Pi Connect

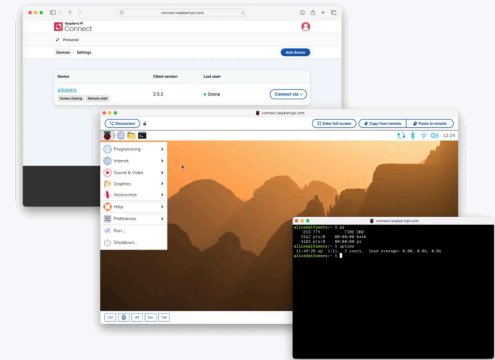
Reach your Pi from anywhere – a handy fallback when local networks are difficult.

- **Raspberry Pi Connect** reaches your Pi from **anywhere**, not just home Wi-Fi.
- Get it at raspberrypi.com/software/connect.
- Free; needs an account, and you sign in **on the Pi** once.
- Useful for screen-sharing or SSH when local discovery is blocked.



Access your Raspberry Pi from anywhere

Raspberry Pi Connect gives you free, simple, out-of-the-box access to your Raspberry Pi from anywhere in the world. It is a secure remote access solution for Raspberry Pi OS, allowing you to connect to your Raspberry Pi desktop and command line directly from any browser.

[Sign in](#)[Read the documentation →](#)



LESSON 0 – INITIAL SETUP • THE COURSE CODE

Download the course code

You need the code in **two places** – on the Pi and on your computer.

- On the **Pi** (to drive the car) **and** on your **computer** (vision lessons 6-8 & 11).
- Get it with `git` – the same command everywhere.
- **Use the HTTPS link** – it just works, no SSH keys to set up. (SSH is only for advanced users.)
- This creates a `Smart-Car-Level-3` folder with all the lesson code.
- No git? Download the **ZIP** from the repo page and unzip it.

any terminal

HTTPS – use this one 

```
git clone https://github.com/renashall/Smart-Car-Level-3.git
```

SSH (advanced – needs SSH keys set up)

```
git clone git@github.com:renashall/Smart-Car-Level-3.git
```

```
cd Smart-Car-Level-3
```



LESSON 0 – INITIAL SETUP • THE COURSE CODE

Same command, three terminals

`git clone` is identical everywhere – only the terminal differs.



Windows

Open **PowerShell** or Windows Terminal. No git?
`winget install Git.Git` or `git-scm.com`.



macOS

Open **Terminal**. Run `git --version` to install the tools, or `brew install git`.



Raspberry Pi

Open **Terminal** or your SSH session. Usually preinstalled; if not, `sudo apt install git`.



LESSON 0 – INITIAL SETUP • INSTALL ON THE PI

Run setup, part 1 (on the Pi)

From the `Code` folder, make the scripts runnable and start part 1.

- It enables Pi **interfaces**, makes `python3` the default `python`, and installs basic **I2C** support.
- It **auto-detects** the Pi model and OS.
- When it finishes, it asks you to **reboot**. These scripts run on the **Pi only**.

the Pi • terminal

```
cd Smart-Car-Level-3/Code
chmod +x setupPart1.sh setupPart2.sh
./setupPart1.sh
```

```
# when it finishes, it reboots
```



LESSON 0 – INITIAL SETUP • INSTALL ON THE PI

Run setup, part 2 (on the Pi)

After the reboot, return to the `Code` folder and run part 2.

- It sets up the **camera** boot config.
- Installs the **LED (WS281x) driver** and the **course Python packages**.
- Then it asks you to **reboot** one more time.

```
the Pi · after the reboot

cd Smart-Car-Level-3/Code
./setupPart2.sh

# two scripts, two reboots – by design
```



LESSON 0 – INITIAL SETUP • INSTALL ON THE PI

What the setup scripts did

You don't need to memorize this – but here's what got configured.

Pi interfaces

I2C, camera, etc.

`python3` default

opens as `python`

WS281x LED driver

the LED strip

Course packages

what the lessons import



No venv on the Pi

The scripts install everything system-wide –
that's why the Pi needs no virtual environment.



LESSON 0 – INITIAL SETUP • INSTALL ON THE PI

Where your code lives: Code/User

Put all your course work in one place.

- Keep your work in `Smart-Car-Level-3/Code/User`, separate from the Freenove code.
- Lesson files start with `import car_setup` so the car's modules import cleanly.
- We'll explain `car_setup.py` properly in **Lesson 1**.

```
project layout

Smart-Car-Level-3/
  Code/
    Server/      # Freenove car code
    Client/      # Freenove desktop app
    User/        # ← your lessons go here
```



LESSON 0 – INITIAL SETUP • INSTALL ON THE PI

Editing code on the Pi: Thonny

A friendly Python editor that's already on the Pi.

- **Thonny** comes with Raspberry Pi OS – a beginner-friendly Python editor, nothing to download.
- Open it on the Pi desktop (over **VNC**): **Menu** → **Programming** → **Thonny**.
- Use it to open, write, and **save** your lesson files in `Smart-Car-Level-3/Code/User`.
- You **run** lessons from the terminal, not Thonny's Run button – more in **Lesson 1**.



Edit & save in Thonny → run in the terminal.



LESSON 0 – INITIAL SETUP • YOUR COMPUTER

Is Python installed? (your computer)

The Pi already has Python – these steps are for your computer's vision lessons.

- The setup scripts handled Python on the **Pi** – this is for your **computer**.
- Open a terminal and check the version.
- See `Python 3.12.x` ? You're set – skip to making a venv. The course needs **3.9+**.

your computer • terminal

```
# Windows  
py --version
```

```
# macOS / Linux  
python3 --version
```



LESSON 0 – INITIAL SETUP • YOUR COMPUTER

Install Python if needed

If the check failed, install Python 3 for your OS.



Windows

Download from python.org/downloads, run it, and tick **"Add python.exe to PATH"**. Or `winget install Python.Python.3.12`.



macOS

Download from python.org/downloads, or with Homebrew: `brew install python`.



Windows' #1 trap: forgetting **"Add to PATH"**. Then close and re-open the terminal and re-check.



LESSON 0 – INITIAL SETUP • YOUR COMPUTER

Create a virtual environment

A venv is a private package box for this project – installs stay here.

- Installs stay in the venv, not on your **whole computer**.
- Make one named `.venv` in the **repo root** – do this once.
- That's also where `Code/setup.py` puts it. You'll reuse it every lesson.

```
your computer · repo root
```

```
cd Smart-Car-Level-3
```

```
# Windows
```

```
py -m venv .venv
```

```
# macOS / Linux
```

```
python3 -m venv .venv
```



LESSON 0 – INITIAL SETUP • YOUR COMPUTER

Activate the venv every time

Activate it whenever you open a terminal to run a lesson.

- Your prompt shows `(.venv)` when it's active – that's your proof.
- Install the packages **once** with `pip install -r requirements.txt`.
- Run `deactivate` when you're done.
- PowerShell blocks the script? Run once: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`.

```
your computer · repo root

# Windows (PowerShell)
.\.venv\Scripts\Activate.ps1

# macOS / Linux
source .venv/bin/activate

# then, once:
pip install -r requirements.txt
```



LESSON 0 – INITIAL SETUP • YOUR COMPUTER

Shortcut: let Code/setup.py do it

Don't want to do slides 39-40 by hand? Run the helper from the repo root.

- It **creates** `.venv` and **installs all the packages** for you.
- No admin or `sudo` needed. `--force` rebuilds it.
- You **still activate** the venv (slide 40) before running a lesson.

your computer · repo root

```
# Windows
python Code\setup.py

# macOS / Linux
python3 Code/setup.py
```



LESSON 0 – INITIAL SETUP • TRY IT YOURSELF

Verify your setup

Confirm each piece works — all six and you're ready for Lesson 1!

- 1 The Pi **boots** and joins Wi-Fi.
- 2 You can **SSH** in: `ssh pi@mypi.local`.
- 3 **VNC Viewer** shows the Pi desktop.
- 4 The `Smart-Car-Level-3` folder exists and **both scripts ran**.
- 5 On your computer the `.venv` activates and `import cv2` runs cleanly.
- 6 You can `cd Smart-Car-Level-3/Code/User`.



The venv `import cv2` check catches missing packages early — verify it now.



LESSON 0 – INITIAL SETUP

Common issues

If something's off, it's almost always one of these.

- ⚠️ **Won't boot / lightning icon** → wrong or weak power adapter (slide 16).
- ❌ **SSH refused / times out** → SSH wasn't enabled in Imager; re-flash.
- ❌ `git: command not found` → install git (slide 32) or use the ZIP.
- ⚠️ **Venv won't activate** → `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`.
- ⚠️ **Can't find the Pi** → same Wi-Fi? Try the IP instead of `.local`.
- ⚠️ **VNC won't connect** → not enabled (slide 27); black screen → set a resolution.
- ❌ **python not found (Windows)** → use `py`, or reinstall with "Add to PATH".
- ❌ `ModuleNotFoundError` → activate `.venv`, then `pip install -r requirements.txt`.
- ❌ **Setup script error (Pi)** → re-run it, read the **last** error line, and check the Pi's internet.



LESSON 0 – INITIAL SETUP

Your robot's brain is alive

Setup is the hardest logistics day – from here on it's coding and fun.

- ✓ Inventoried the kit and learned **battery safety**
- ✓ **Flashed** the OS and reached the Pi with **SSH + VNC**
- ✓ Installed the **course code** on the Pi and your computer

UP NEXT

Lesson 1 Your first program

Write your first program and light up the car's components.



• COACH ONLY • HIDDEN SLIDE

End-of-class checklist

Don't present this slide. Run through it right after class.

1

Post the after-class recap

Summarise what you covered on the class page so students can catch up.

2

Log absent students

Post anyone who missed class in the **#attendance** Discord channel.

3

Charge for next class

Charge the cars before Lesson 1 – low battery breaks the LED and servo demos.

EVERY CLASS

✓ Post a class announcement

✓ Introductions if needed

✓ Note absences in #attendance



• COACH ONLY • HIDDEN SLIDE

Coach notes & checklist

Don't present this slide. This session often spans more than one class – don't rush it.

1 Before class – have ready

Charged spare 18650s, a known-good adapter per Pi model, a spare **pre-flashed SD card**, the Wi-Fi name/password, and the repo link (also in Discord / class page). Pre-flash a demo card to show the Imager steps live.

2 Lead safety seriously

Battery handling (slide 14) and the correct Pi power adapter (slide 16) – under-power causes most boot issues.

3 Pi vs. computer

The Pi uses system Python (**no venv**); the venv steps 37-41 are computer-only, for the vision lessons.

4 Record-keeping & assets

Write down each student's **hostname / username / password**. Still to source: Imager main screen (slide 21) and the OS-customisation dialog with SSH enabled (slide 22).

BIGGEST TIME-SINKS

SSH not enabled in Imager Step 3

Wrong / weak power adapter

`.local` not resolving on Windows

Windows Python without "Add to PATH"